AUDIT REPORT

supply-chain-twin: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/supply-chain-twin
Numbers taken from commit	`a4f50f5289cc8c298a8e842f8aadd139c609579c` (outputs last changed in `a4f50f5 (2026-10-13 19:09:20 +0400)`)
Video	`docs/demo/demo.mp4` · 92.8 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 18:46
Automated checks	67 pass · 0 fail · 1 manual (of 68 figures)
Assumptions	10 (A-01 to A-10)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                    # data/ + outputs/*.csv|json (about 2.5 min on 4 cores)
python3 docs/demo/source/tools/export_search_walk.py docs/demo/source/assets/search_walk.json
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_change_sheet.py
docs/demo/source/assets/change_sheet.json
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/search_walk.json
docs/demo/source/compositions/s04-running.html --name WALK
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/change_sheet.json
docs/demo/source/compositions/value.html --name SHEET
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q
looks -o renders/supply-chain-twin-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/supply-chain-twin-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt
yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks,
MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 industry chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem', 'Why it is hard'	README lines 9-31	-	general knowledge	✓ verified
F-03	2 DCs	02, 05, 10	data/mms_locations.csv	distinct dc_id	-	direct	✓ verified
F-04	36 stores	02, 05, 08, 10	outputs/data_quality.json	stores	-	direct	✓ verified
F-05	26 trucks	02, 10	outputs/data_quality.json	trucks_dispatchable	28 in source - 2 not dispatchable	direct	✓ verified
F-06	3.1M order lines	02, 05	outputs/data_quality.json	order_lines = 3,126,764	3,126,764 / 1e6, 1 decimal	derived	✓ verified
F-07	2 years	02, 05	src/supply_twin/calendar.py:12	HISTORY_DAYS = 731 (2024-01-01..2025-12-31)	-	direct	✓ verified
F-08	120 SKUs in 8 categories	02, 05	data/mms_items.csv; outputs/data_quality.json	skus; distinct category	-	direct	✓ verified
F-09	91-day quarter × 30 paired seeds	02	src/supply_twin/sim.py:60; outputs/strategy_table.csv	horizon_days = 91; n_seeds	-	direct	✓ verified
F-10	the twin's demand runs 3.3% low / −3.3% demand bias	02, 04	outputs/fidelity.json	bias.twin = -0.0327	x 100, 1 decimal	direct	✓ verified
F-11	SOURCE DATA → DIGITAL TWIN → PROPOSE → SIMULATE → FAIR TEST → DECISION	03	src/supply_twin/	load_sources, build_twin, NeighbourProposer.propose, simulate, Evaluator.compare, search	-	direct	✓ verified
F-12	✓ cheaper than today ✓ re-tested on new days ✓ weak spots shown	03	outputs/search_neighbour_promo.json; outputs/search_neighbour_promo_robustness.csv	holdout.cost_diff < 0; holdout.n_seeds = 30; any robustness row not accepted	-	derived	✓ verified
F-13	until nothing wins	03	outputs/search_neighbour_promo.csv; agent.py:118-119	last round has 0 accepted candidates	-	direct	✓ verified
F-14	Less stock cover (crosses 0) · Promotion-aware ordering (below 0, above floor) · Two fewer trucks (clears floor)	04	outputs/search_neighbour_promo.csv	round 1 cover=0.5d: +0.139 (-0.078..+0.355); round 3 promo_aware=yes: -0.516 (-0.785..-0.247); round 2 trucks-2: -2.169 (-2.385..-1.954)	cost_diff{,_low,_high} / baseline_cost x 100, 3 decimals; must equal the IV array in compositions/explainer.html	derived	✓ verified
F-15	✗ could be noise · ✗ real, but inside twin error · ✓ clears the floor	04	outputs/search_neighbour_promo.csv	accepted, p_value vs alpha	A: p >= alpha, not accepted; B: p < alpha, not accepted, fill ok; C: accepted	derived	✓ verified
F-16	Floor = 0.5% of quarterly cost / cost CI < −0.5% / floor −0.5%	04, 05, 06	src/supply_twin/harness.py:34	MIN_COST_GAIN_SHARE = 0.005	-	direct	✓ verified
F-17	twin.py · TwinState · fidelity.py · sim.py · simulate() · harness.py · llm.py · agent.py · cli.py · load_sources()	05	src/supply_twin/	classes and functions	-	direct	✓ verified
F-18	28 trucks	05	outputs/data_quality.json	trucks_in_source	-	direct	✓ verified
F-19	4,320 store-SKU rates	05, 08	outputs/data_quality.json	stores x skus	36 x 120 = 4,320	derived	✓ verified
F-20	Data quality · data_quality.json	05	src/supply_twin/cli.py:49	_write_json(out / 'data_quality.json')	-	direct	✓ verified
F-21	load_sources() · 4 CSV files	05	src/supply_twin/twin.py:74-80	four read_csv calls	-	direct	✓ verified
F-22	Backtest · 13 weeks	05	outputs/fidelity.json	holdout_days = 91	91 / 7 = 13	derived	✓ verified
F-23	WAPE 17.8% / 17.8%	05, 08	outputs/fidelity.json	wape_store_sku_week.twin	x 100	direct	✓ verified
F-24	σ = 0.104 demand error	05	outputs/fidelity.json	rate_error_sigma	-	direct	✓ verified
F-25	bias −3.3%	05	outputs/fidelity.json	bias.twin	x 100	direct	✓ verified
F-26	91 days · sell 21:00 · trucks 05–19 h	05	src/supply_twin/sim.py:27-28, :60	SHIFT_START, SHIFT_END, SALES_HOUR, horizon_days	-	direct	✓ verified
F-27	≈0.1–0.2 s per simulated quarter	05	docs/adr/0001-paired-seeds-and-a-practical-floor-decide-strategies.md:21	'runs in about 0.1-0.2 s'	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-28	1,930 runs	05	outputs/search_neighbour_promo.json	simulation_runs_total (Evaluator.runs: unique runs executed, including the evaluation table)	-	direct	✓ verified
F-29	pool ≤ 4	05	src/supply_twin/harness.py:107	min(4, os.cpu_count())	-	direct	✓ verified
F-30	180 s timeout	05	src/supply_twin/llm.py:25	timeout: float = 180	-	direct	✓ verified
F-31	accept if cost CI < −0.5% and fill loss < 0.2 pts	05	src/supply_twin/harness.py:29, :34, :76-86	MIN_COST_GAIN_SHARE, FILL_NON_INFERIORITY	-	direct	✓ verified
F-32	30 hold-out seeds × 3 scenarios	05	outputs/search_neighbour_promo.json; outputs/search_neighbour_promo_robustness.csv	holdout.n_seeds; robustness rows	-	direct	✓ verified
F-33	round 8 · 71 tested (counts up through rounds 1-8)	05	outputs/search_neighbour_promo.csv / .json	rounds, candidates_tested, cumulative per round	cumulative tested per round = 8, 17, 26, 35, 44, 53, 62, 71	direct	✓ verified
F-34	make run ≈ 2.5 min / ~2.5 min	05, 10	outputs/run_log.txt	'2:26.26 total'	146.26 s / 60 = 2.44 min, rounded to the nearest half minute	derived	✓ verified
F-35	3 scenarios · base · promo · outage	05	src/supply_twin/sim.py:71-75	SCENARIOS keys base, promo, fleet_outage	-	direct	✓ verified
F-36	71 candidates over 8 rounds; ▲ = 6 far costlier ones	06	outputs/search_neighbour_promo.csv → demo assets/search_walk.json, injected into compositions/s04-running.html	round, cost_diff, accepted, chosen	dot y = cost_diff / 1000 (k); off-chart if > 320k	direct	✓ verified
F-37	1 every 2 days −143,755 · 2 trucks −2 −50,217 · 3 trucks −4 −56,085 · 4 trucks −6 −44,795 · 5 trucks −8 −46,004 · 6 promo-aware −20,339 · 7 safety z 0.84 −49,375 · 8 nothing passes	06	outputs/search_neighbour_promo.csv	rows with chosen = True	rounded to whole units	direct	✓ verified
F-38	2,458,533 (start of the counter)	06	outputs/search_neighbour_promo.csv	round 1 baseline_cost	-	direct	✓ verified
F-39	2,047,964	06	outputs/search_neighbour_promo.csv	round 8 baseline_cost	-	direct	✓ verified
F-40	20 paired seeds per test	06	outputs/search_neighbour_promo.json	search_seeds	-	direct	✓ verified
F-41	2,458,347 → 2,051,146	06, 08	outputs/search_neighbour_promo.json	holdout.baseline_cost, holdout.cost_diff	2,458,346.6 - 407,200.4 = 2,051,146.1	derived	✓ verified
F-42	−16.6%	06, 08	outputs/search_neighbour_promo.json	holdout.cost_diff / holdout.baseline_cost	-407,200.4 / 2,458,346.6 x 100 = -16.56%	derived	✓ verified
F-43	fill rate −0.13 pts	06	outputs/search_neighbour_promo.json	holdout.fill_diff_pts	-	direct	✓ verified
F-44	truck use +12.9 pts	06	outputs/search_neighbour_promo.json	holdout.util_diff_pts	-	direct	✓ verified
F-45	searched plan: order every 2 days · safety z 0.84 · 8 fewer trucks · promo-aware	06	outputs/search_neighbour_promo.json	final (start = z=1.28 review=1d cover=1d trucks+0 promo_aware=no)	review=2d → order every 2 days; z=0.84 → safety z 0.84; trucks-8 → 8 fewer trucks; promo_aware=yes → promo-aware	direct	✓ verified
F-46	Current policy 2,460,080 … High service 2,616,577 (+156,497 ±2,984)	07	demo assets/change_sheet.json (re-run of the repo's evaluation, seeds 1000-1029), injected into compositions/value.html; equals outputs/strategy_table.csv scenario=promo	total_cost, cost_vs_current, cost_vs_current_ci_half, fill_rate, promo_fill_rate, beats_current	-	direct	✓ verified
F-47	3 of 6 changes clear the gate	07	outputs/strategy_table.csv	scenario=promo, beats_current	-	direct	✓ verified
F-48	best: order every 2 days, −142,008 a quarter	07	outputs/strategy_table.csv	scenario=promo, strategy='order every 2 days', cost_vs_current	-	direct	✓ verified
F-49	65% → 78%	07	outputs/strategy_table.csv	scenario=promo, load_fill (0.6461, 0.7760)	x 100, whole percent	direct	✓ verified
F-50	98.4% → 96.8%	07	outputs/strategy_table.csv	scenario=promo, promo_fill_rate (0.9844, 0.9679)	x 100, 1 decimal	direct	✓ verified
F-51	PROMO QUARTER · 30 SEEDS / 30 seeds	07	demo assets/change_sheet.json	n_seeds	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-52	DCs → 36 stores, deliveries on alternating days	07	data/mms_locations.csv; src/supply_twin/sim.py:291 ((day + s) % review_days)	store_id, dc_id, distance_km	radius = 26 px + km x 0.68 px; angle illustrative	direct	✓ verified
F-53	−5.8%	08	outputs/strategy_table.csv	scenario=promo, total_cost: 2,318,072 vs 2,460,080	(2,318,072 / 2,460,080 - 1) x 100 = -5.77%	derived	✓ verified
F-54	2,460,080 → 2,318,072	08	outputs/strategy_table.csv	scenario=promo, total_cost	-	direct	✓ verified
F-55	naive: 24.5% · 29.1%	08	outputs/fidelity.json	wape_store_sku_week.recent_8wk_mean, same_day_last_year	x 100	direct	✓ verified
F-56	99.95% · current policy 98.44%	08	outputs/strategy_table.csv	scenario=promo, promo_fill_rate	x 100, 2 decimals	direct	✓ verified
F-57	Promo quarter · tuned on: −16.6% cost · −0.13 · TRAP	09	outputs/search_neighbour_promo_robustness.csv	scenario=promo: fill_diff_pts, cost_diff / baseline_cost, accepted	-407,200 / 2,458,347 x 100 = -16.6%	derived	✓ verified
F-58	Normal quarter · cost −398,970 · −0.21 · FAILS	09	outputs/search_neighbour_promo_robustness.csv	scenario=base	-	direct	✓ verified
F-59	Fleet 20% down · cost +63,322 · −1.88 · FAILS · 1,336 loads a quarter arrive late	09	outputs/search_neighbour_promo_robustness.csv	scenario=fleet_outage	late loads = final_late_loads (mean over 30 hold-out seeds, one 91-day quarter) rounded	direct	✓ verified
F-60	0.2 limit / past 0.2 pts fails	09	src/supply_twin/harness.py:29, :79-80	FILL_NON_INFERIORITY = 0.002 on the lower bound of the fill change	-	direct	✓ verified
F-61	Re-test every plan on all 3 scenarios · The run re-tests the searched plan on every scenario	09	src/supply_twin/cli.py:65-82 (cmd_evaluate), :157-181 (cmd_robustness), :219-222 (called after the search)	for ... in SCENARIOS.items(); cmd_robustness(...)	-	direct	✓ verified
F-65	Promo quarter cost −142,008 −0.08 PASS · Normal quarter cost −169,129 +0.01 PASS · Fleet 20% down cost −200,926 +0.09 PASS	09	outputs/strategy_table.csv	strategy='order every 2 days', scenario in promo/base/fleet_outage: fill_vs_current_pts, cost_vs_current, beats_current	-	direct	✓ verified
F-66	only this change passes all three	09	outputs/strategy_table.csv	beats_current per strategy across the 3 scenarios	strategies with beats_current in every scenario = {order every 2 days}	derived	✓ verified
F-67	saves 142,008 to 200,926 a quarter, fill within the limit	09	outputs/strategy_table.csv	strategy='order every 2 days': cost_vs_current min/max; beats_current all	-max(cost_vs_current) .. -min(cost_vs_current)	derived	✓ verified
F-68	The search still tunes on one scenario. Next: score a weighted mix that includes breakdowns.	09	README 'Known limitations' (first bullet) and 'Insights and learning'	'search against a weighted mix of scenarios'; 'neither is implemented'	-	direct	✓ verified
F-62	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-63	SimPy · pandas · SciPy · Python	10	pyproject.toml dependencies	simpy, pandas, scipy	-	direct	✓ verified
F-64	$ make run	10	Makefile	run target	-	direct	✓ verified

2. Automated check run (2026-10-14 18:46)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 industry chips	-	manual
F-02	true	true	✓ pass
F-03	2	2	✓ pass
F-04	36	36	✓ pass
F-05	26	26	✓ pass
F-06	3.1M	3.1M	✓ pass
F-07	true	true	✓ pass
F-08	true	true	✓ pass
F-09	true	true	✓ pass
F-10	-3.3%	-3.3%	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	true	true	✓ pass
F-14	true	true	✓ pass
F-15	true	true	✓ pass
F-16	true	true	✓ pass
F-17	true	true	✓ pass
F-18	28	28	✓ pass

ID	Expected on slide	Recomputed from repo	Result
F-19	4,320	4,320	✓ pass
F-20	true	true	✓ pass
F-21	true	true	✓ pass
F-22	13	13	✓ pass
F-23	17.8%	17.8%	✓ pass
F-24	0.104	0.104	✓ pass
F-25	-3.3%	-3.3%	✓ pass
F-26	true	true	✓ pass
F-27	true	true	✓ pass
F-28	1,930	1,930	✓ pass
F-29	true	true	✓ pass
F-30	true	true	✓ pass
F-31	true	true	✓ pass
F-32	true	true	✓ pass
F-33	true	true	✓ pass
F-34	2.5 min	2.5 min	✓ pass
F-35	true	true	✓ pass
F-36	true	true	✓ pass
F-37	true	true	✓ pass
F-38	2,458,533	2,458,533	✓ pass
F-39	2,047,964	2,047,964	✓ pass
F-40	20	20	✓ pass
F-41	2,458,347 → 2,051,146	2,458,347 → 2,051,146	✓ pass
F-42	-16.6%	-16.6%	✓ pass
F-43	-0.13	-0.13	✓ pass
F-44	+12.9	+12.9	✓ pass
F-45	z=0.84 review=2d cover=1d trucks-8 promo_aware=yes	z=0.84 review=2d cover=1d trucks-8 promo_aware=yes	✓ pass
F-46	true	true	✓ pass
F-47	3 of 6	3 of 6	✓ pass
F-48	-142,008	-142,008	✓ pass
F-49	65% → 78%	65% → 78%	✓ pass
F-50	98.4% → 96.8%	98.4% → 96.8%	✓ pass
F-51	30	30	✓ pass
F-52	true	true	✓ pass
F-53	-5.8%	-5.8%	✓ pass
F-54	2,460,080 → 2,318,072	2,460,080 → 2,318,072	✓ pass
F-55	24.5% / 29.1%	24.5% / 29.1%	✓ pass
F-56	99.95% / 98.44%	99.95% / 98.44%	✓ pass
F-57	-0.13 -16.6% True	-0.13 -16.6% True	✓ pass
F-58	-0.21 -398,970 False	-0.21 -398,970 False	✓ pass
F-59	-1.88 +63,322 1,336 False	-1.88 +63,322 1,336 False	✓ pass
F-60	true	true	✓ pass
F-61	true	true	✓ pass
F-65	-0.08 -142,008 True / +0.01 -169,129 True / +0.09 -200,926 True	-0.08 -142,008 True / +0.01 -169,129 True / +0.09 -200,926 True	✓ pass
F-66	['order every 2 days']	['order every 2 days']	✓ pass
F-67	142,008 to 200,926 True	142,008 to 200,926 True	✓ pass
F-68	true	true	✓ pass
F-62	true	true	✓ pass
F-63	true	true	✓ pass
F-64	true	true	✓ pass

3. Assumptions register

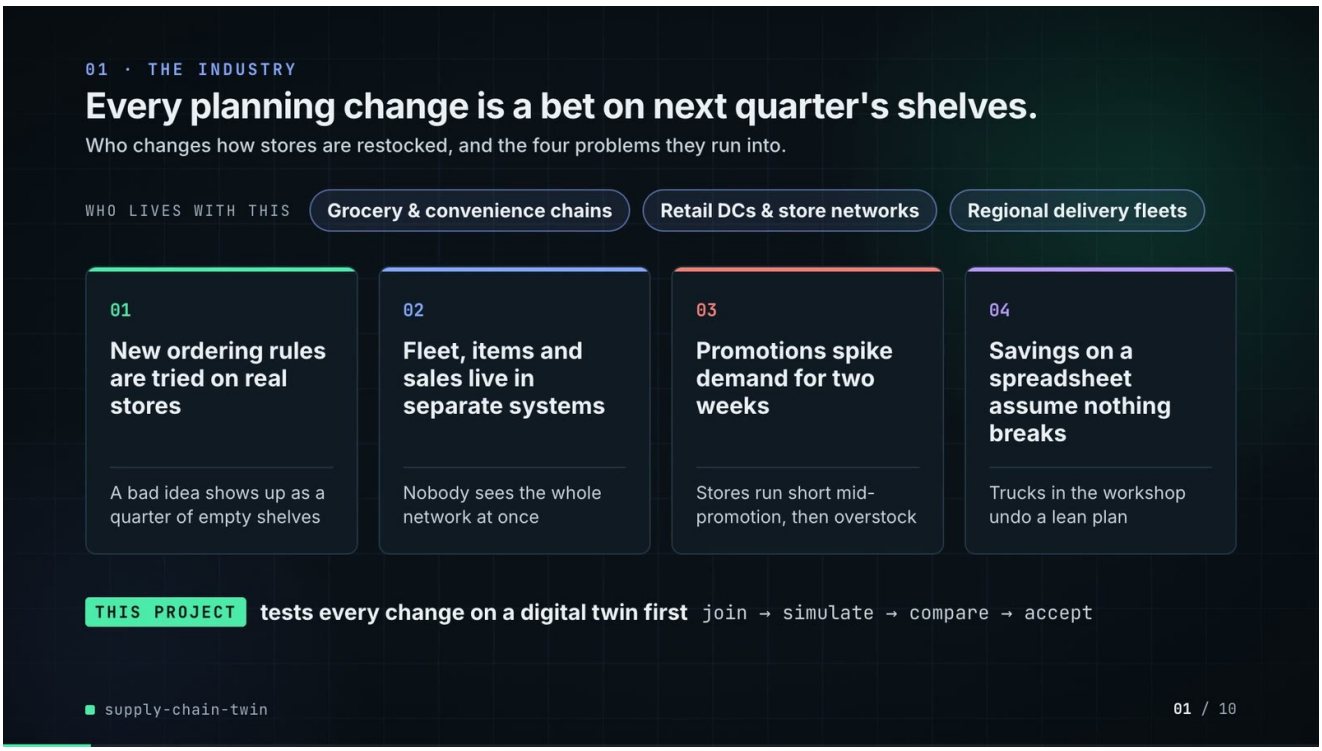
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, written by a seeded generator (seed 7) shaped like the three source systems: 28 trucks, 120 SKUs, 36 stores, 2 DCs, two years of daily order lines with 4% annual growth, holidays and category promotions. The production data it mirrors is proprietary.	scripts/generate_data.py:151 (generate, seed=7), :110 (4% growth); README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Acceptance floor: a change is accepted only if the whole 95% interval of its total-cost change lies below -0.5% of baseline cost. The 0.5% is a judgement set from the twin's -3.3% demand bias, not a statistical result.	src/supply_twin/harness.py:34 (MIN_COST_GAIN_SHARE = 0.005); docs/adr/0001 'Alternatives rejected'	F-14, F-15, F-16, F-31, F-36, F-37, F-39, F-46, F-47, F-48, F-65, F-66, F-67	slides 04, 05, 06, 07, 09
A-03	Fill non-inferiority: a change may lose at most 0.2 points of fill rate (lower bound of the fill-change interval).	src/supply_twin/harness.py:29 (FILL_NON_INFERIORITY = 0.002)	F-31, F-36, F-37, F-46, F-47, F-57, F-58, F-59, F-60, F-65, F-66, F-67	slides 05, 06, 07, 09
A-04	Seeds: the search uses 20 seeds (0-19) with a Bonferroni split of alpha per round; the evaluation table uses 30 seeds (1000-1029); the searched plan is confirmed on 30 hold-out seeds (5000-5029) the search never saw.	src/supply_twin/cli.py:189-191, :207-209; agent.py:110	F-09, F-12, F-32, F-40, F-41, F-42, F-43, F-44, F-51	slides 02, 03, 05, 06, 07, 08

ID	Assumption	Source	Figures affected	Slides affected
A-05	Cost constants are stated, not estimated: holding 25% a year, backroom overflow 1.0 per case-day, driver 28 per hour, trucks at 50 km/h, fleet daily fixed cost from the fleet file. Changing the overflow price changes which review period wins.	src/supply_twin/sim.py:29-34; README 'Known limitations'	F-14, F-36, F-37, F-39, F-41, F-42, F-46, F-48, F-53, F-54, F-57, F-58, F-59	slides 04, 06, 07, 08, 09
A-06	Scenarios: a 91-day horizon; the promotion quarter puts beverages and snacks on a 14-day promotion from day 35; the outage scenario takes 20% of each DC's trucks out of service for the whole quarter.	src/supply_twin/sim.py:60-64, :73-74	F-09, F-26, F-35, F-59, F-61, F-65, F-66	slides 02, 05, 09
A-07	Simulated demand carries the store-SKU error spread measured in the 13-week backtest (sigma 0.104), so the simulated customers are as unpredictable as the history showed.	src/supply_twin/fidelity.py:37 (holdout_days=91), :80-89 (rate_error_sigma); cli.py:202-204	F-10, F-23, F-24, F-25, F-56	slides 02, 04, 05, 08
A-08	The neighbour proposer moves one lever one grid step at a time (z, review days, cover days, fleet size, promotion awareness); the search stops when a round accepts nothing (here after 8 rounds).	src/supply_twin/agent.py:23-29 (GRID), :49-63 (NeighbourProposer), :105-121 (search)	F-13, F-33, F-36, F-37, F-45	slides 03, 05, 06
A-09	Video rounding: money to whole units with thousands separators, percentages to 1 decimal (fill rates to 2), points to 2 decimals; '3.1M', '~2.5 min' and '≈0.1-0.2 s' are rounded labels.	this audit (display convention)	F-06, F-10, F-14, F-24, F-34, F-42, F-49, F-50, F-53, F-56	slides 02, 04, 05, 06, 07, 08, 10
A-10	Slide 01 (industry, everyday problems) is general retail-planning knowledge used as context. It is not a result of this repo and carries no numbers.	author's framing; README 'The problem' supports problems 1, 2 and 4	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every planning change is a bet on next quarter's shelves.

Who changes how stores are restocked, and the four problems they run into.

WHO LIVES WITH THIS · Grocery & convenience chains · Retail DCs & store networks · Regional delivery fleets

01 New ordering rules are tried on real stores — A bad idea shows up as a quarter of empty shelves

02 Fleet, items and sales live in separate systems — Nobody sees the whole network at once

03 Promotions spike demand for two weeks — Stores run short mid-promotion, then overstock

04 Savings on a spreadsheet assume nothing breaks — Trucks in the workshop undo a lean plan

THIS PROJECT tests every change on a digital twin first · join → simulate → compare → accept

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 industry chips	`author's framing · -`	-	manual
F-02	4 problem cards	`README 'The problem', 'Why it is hard' ·` README lines 9-31	-	✓

Assumptions: A-10

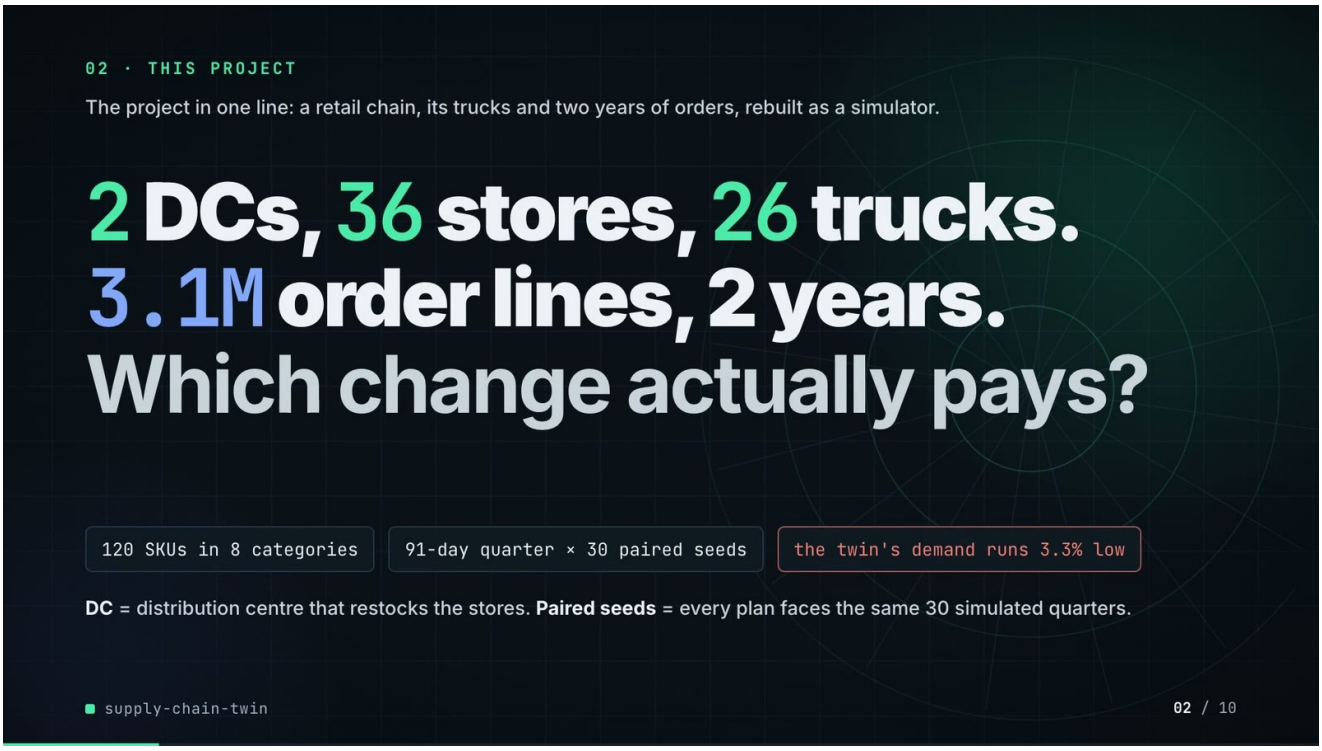
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-12.0 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

The project in one line: a retail chain, its trucks and two years of orders, rebuilt as a simulator.

2 DCs, 36 stores, 26 trucks. 3.1M order lines, 2 years. Which change actually pays?

120 SKUs in 8 categories · 91-day quarter × 30 paired seeds · the twin's demand runs 3.3% low

DC = distribution centre that restocks the stores. Paired seeds = every plan faces the same 30 simulated quarters.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	2 DCs	`data/mms_locations.csv` · distinct dc_id	-	✓
F-04	36 stores	`outputs/data_quality.json` · stores	-	✓
F-05	26 trucks	`outputs/data_quality.json` · trucks_dispatchable	28 in source - 2 not dispatchable	✓
F-06	3.1M order lines	`outputs/data_quality.json` · order_lines = 3,126,764	3,126,764 / 1e6, 1 decimal	✓
F-07	2 years	`src/supply_twin/calendar.py:12` · HISTORY_DAYS = 731 (2024-01-01..2025-12-31)	-	✓
F-08	120 SKUs in 8 categories	`data/mms_items.csv`; `outputs/data_quality.json` · skus; distinct category	-	✓
F-09	91-day quarter × 30 paired seeds	`src/supply_twin/sim.py:60`; `outputs/strategy_table.csv` · horizon_days = 91; n_seeds	-	✓
F-10	the twin's demand runs 3.3% low / −3.3% demand bias	`outputs/fidelity.json` · bias.twin = -0.0327	x 100, 1 decimal	✓

Assumptions: A-01, A-04, A-06, A-07

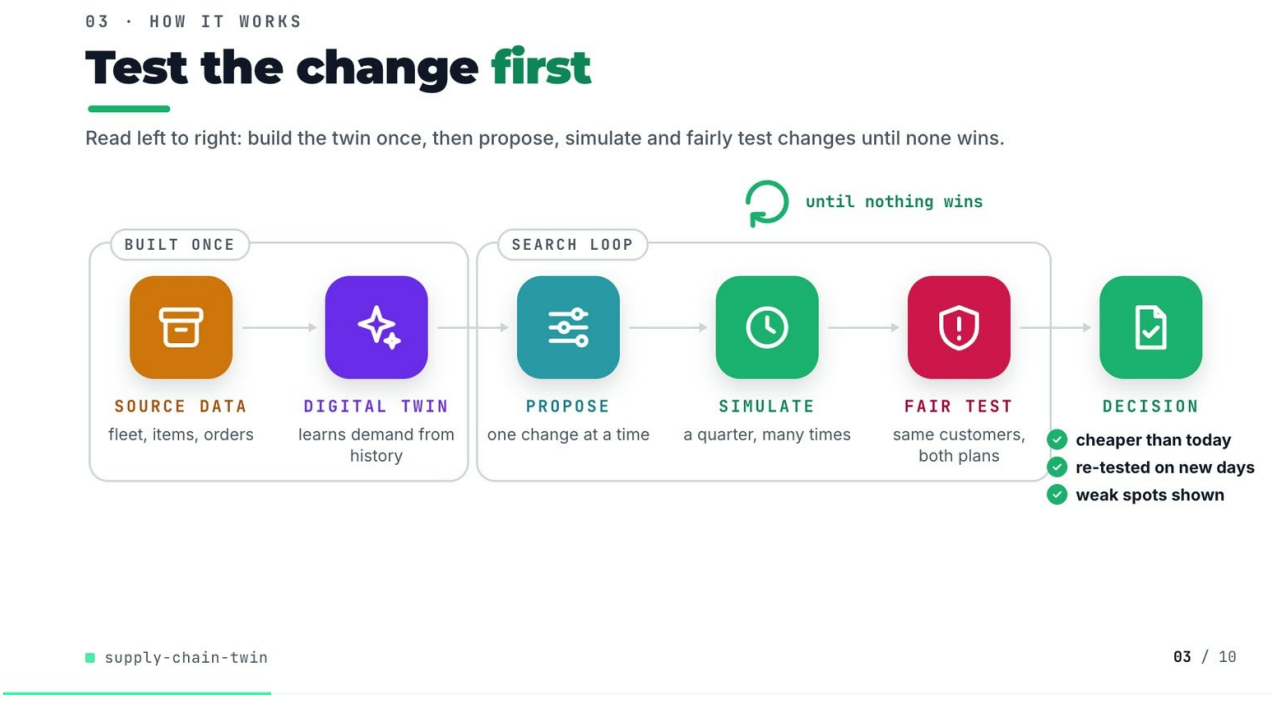
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; counts come from outputs/data_quality.json.

Slide 03: How it works (white chapter, slide A)

11.7-19.7 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

Test the change first

Read left to right: build the twin once, then propose, simulate and fairly test changes until none wins.

[BUILT ONCE: SOURCE DATA fleet, items, orders → DIGITAL TWIN learns demand from history] → [SEARCH LOOP: PROPOSE one change at a time → SIMULATE a quarter, many times → FAIR TEST same customers, both plans] → DECISION ✓ cheaper than today ✓ re-tested on new days ✓ weak spots shown

until nothing wins (loop over SIMULATE)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-11	SOURCE DATA → DIGITAL TWIN → PROPOSE → SIMULATE → FAIR TEST → DECISION	`src/supply_twin/` · load_sources, build_twin, NeighbourProposer.propose, simulate, Evaluator.compare, search	-	✓
F-12	✓ cheaper than today ✓ re-tested on new days ✓ weak spots shown	`outputs/search_neighbour_promo.json`; `outputs/search_neighbour_promo_robustness.csv` · holdout.cost_diff < 0; holdout.n_seeds = 30; any robustness row not accepted	-	✓
F-13	until nothing wins	`outputs/search_neighbour_promo.csv`; `agent.py:118-119` · last round has 0 accepted candidates	-	✓

Assumptions: A-04, A-08

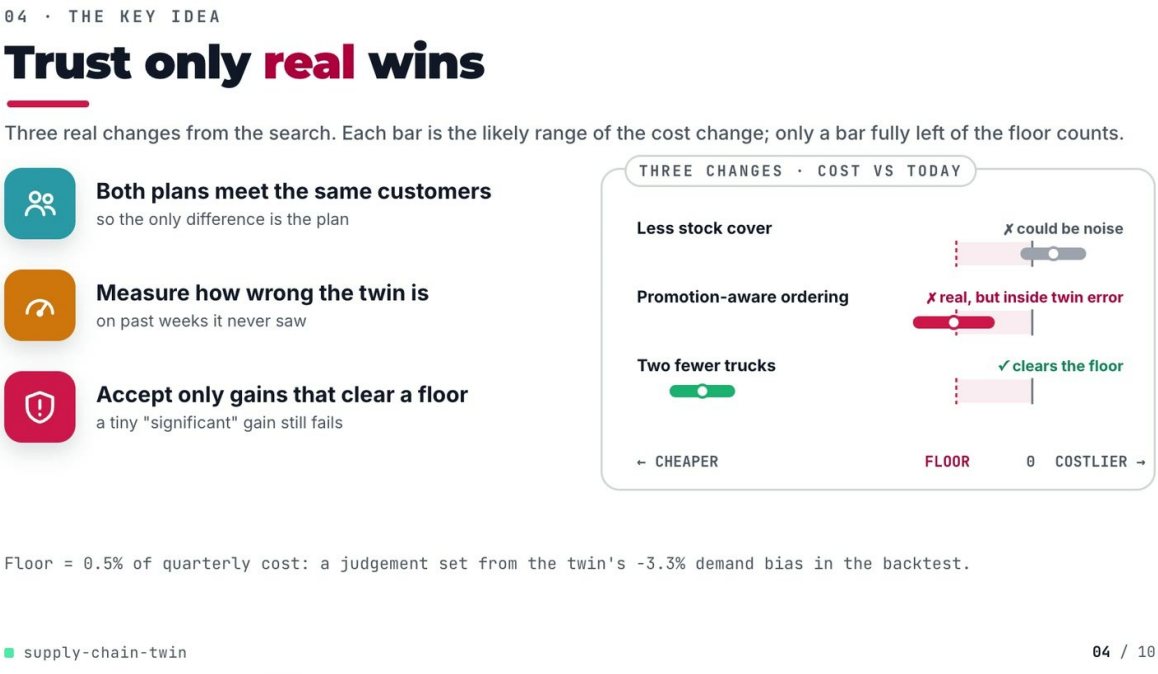
Illustrative, not data:

- Icons and colours are a visual language, not data.
- 'new days' means new random seeds, i.e. new simulated customers.

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth comes from the search JSON and robustness CSV (F-12).

Slide 04: The key idea (white chapter, slide B)

19.7-27.7 s · composition `compositions/explainer.html (#ex-b ..., IV array)`



On-screen text (verbatim)

04 · THE KEY IDEA

Trust only real wins

Three real changes from the search. Each bar is the likely range of the cost change; only a bar fully left of the floor counts.

Both plans meet the same customers — so the only difference is the plan

Measure how wrong the twin is — on past weeks it never saw

Accept only gains that clear a floor — a tiny "significant" gain still fails

THREE CHANGES · COST VS TODAY: Less stock cover ✗ could be noise · Promotion-aware ordering ✗ real, but inside twin error · Two fewer trucks ✓ clears the floor · ← CHEAPER · FLOOR · 0 · COSTLIER →

Floor = 0.5% of quarterly cost: a judgement set from the twin's −3.3% demand bias in the backtest.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-10	the twin's demand runs 3.3% low / −3.3% demand bias	`outputs/fidelity.json` · bias.twin = -0.0327	x 100, 1 decimal	✓
F-14	Less stock cover (crosses 0) · Promotion-aware ordering (below 0, above floor) · Two fewer trucks (clears floor)	`outputs/search_neighbour_promo.csv` · round 1 cover=0.5d: +0.139 (-0.078..+0.355); round 3 promo_aware=yes: -0.516 (-0.785..-0.247); round 2 trucks-2: -2.169 (-2.385..-1.954)	cost_diff{,_low,_high} / baseline_cost x 100, 3 decimals; must equal the IV array in compositions/explainer.html	✓
F-15	✗ could be noise · ✗ real, but inside twin error · ✓ clears the floor	`outputs/search_neighbour_promo.csv` · accepted, p_value vs alpha	A: p >= alpha, not accepted; B: p < alpha, not accepted, fill ok; C: accepted	✓
F-16	Floor = 0.5% of quarterly cost / cost CI < −0.5% / floor −0.5%	`src/supply_twin/harness.py:34` · MIN_COST_GAIN_SHARE = 0.005	-	✓

Assumptions: A-02, A-03, A-07

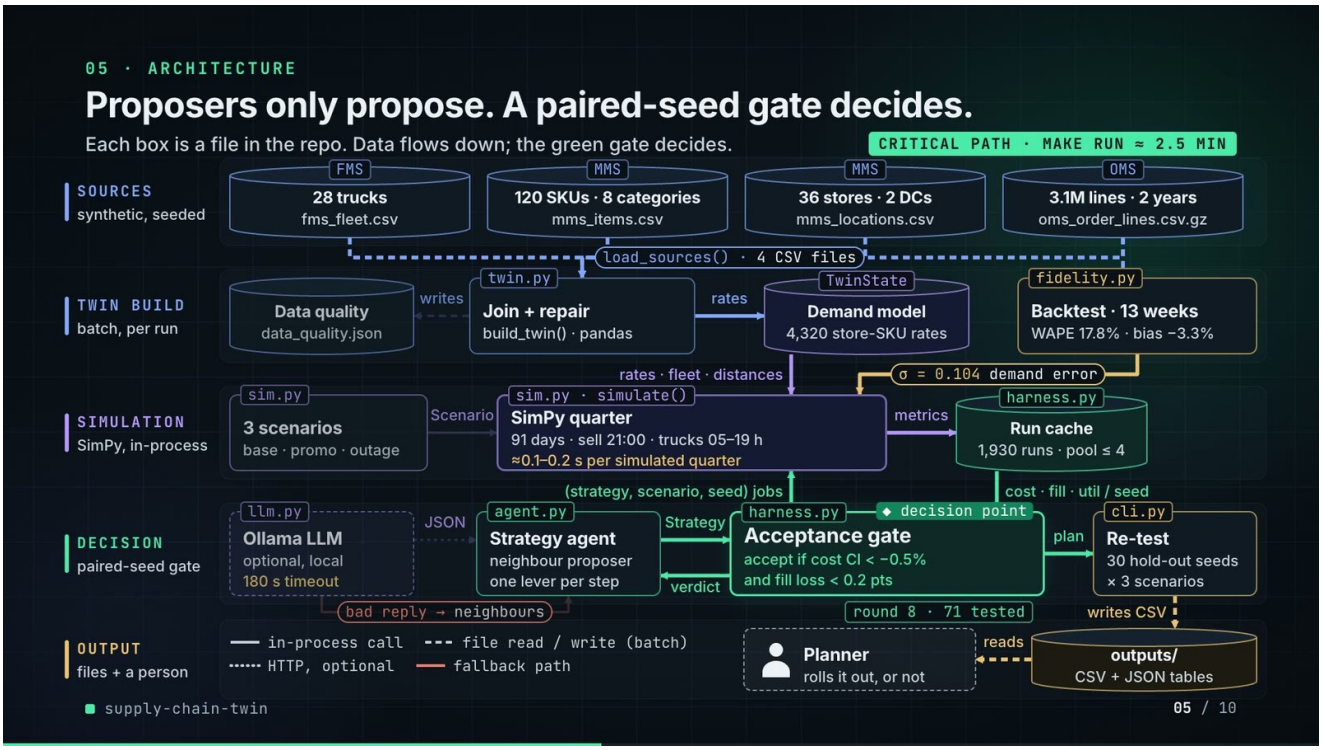
Illustrative, not data:

- The three intervals are real candidates from the search log (F-14), drawn to scale on a -2.6%..+0.6% axis; 'today' in the pill means the plan in hand at that round, not always the original policy.

To change this slide: Markup under #ex-b and the IV array in compositions/explainer.html; F-14 checks the IV values against the CSV.

Slide 05: Architecture

27.4-43.4 s · composition `compositions/s03-architecture.html (BANDS, NODES, WIRES, LABELS)`



On-screen text (verbatim)

05 · ARCHITECTURE

Proposers only propose. A paired-seed gate decides.

Each box is a file in the repo. Data flows down; the green gate decides.

SOURCES (synthetic, seeded): FMS 28 trucks fms_fleet.csv · MMS 120 SKUs · 8 categories mms_items.csv · MMS 36 stores · 2 DCs mms_locations.csv · OMS 3.1M lines · 2 years oms_order_lines.csv.gz → load_sources() · 4 CSV files

TWIN BUILD (batch, per run): twin.py Join + repair build_twin() · pandas → writes Data quality data_quality.json; → rates TwinState Demand model 4,320 store-SKU rates; fidelity.py Backtest · 13 weeks WAPE 17.8% · bias −3.3%

SIMULATION (SimPy, in-process): sim.py 3 scenarios base · promo · outage → Scenario → sim.py · simulate() SimPy quarter 91 days · sell 21:00 · trucks 05–19 h ≈0.1–0.2 s per simulated quarter → metrics → harness.py Run cache 1,930 runs · pool ≤ 4; rates · fleet · distances; σ = 0.104 demand error

DECISION (paired-seed gate): llm.py Ollama LLM optional, local 180 s timeout → JSON → agent.py Strategy agent neighbour proposer, one lever per step → Strategy → harness.py Acceptance gate ◆ decision point accept if cost CI < −0.5% and fill loss < 0.2 pts → verdict; (strategy, scenario, seed) jobs; cost · fill · util / seed; bad reply → neighbours; round 8 · 71 tested → plan → cli.py Re-test 30 hold-out seeds × 3 scenarios

OUTPUT (files + a person): writes CSV → outputs/ CSV + JSON tables → reads → Planner rolls it out, or not

Legend: in-process call · file read / write (batch) · HTTP, optional · fallback path · CRITICAL PATH · MAKE RUN ≈ 2.5 MIN

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	2 DCs	`data/mms_locations.csv` · distinct dc_id	-	✓
F-04	36 stores	`outputs/data_quality.json` · stores	-	✓
F-06	3.1M order lines	`outputs/data_quality.json` · order_lines = 3,126,764	3,126,764 / 1e6, 1 decimal	✓
F-07	2 years	`src/supply_twin/calendar.py:12` · HISTORY_DAYS = 731 (2024-01-01..2025-12-31)	-	✓
F-08	120 SKUs in 8 categories	`data/mms_items.csv`; `outputs/data_quality.json` · skus; distinct category	-	✓
F-16	Floor = 0.5% of quarterly cost / cost CI < −0.5% / floor −0.5%	`src/supply_twin/harness.py:34` · MIN_COST_GAIN_SHARE = 0.005	-	✓
F-17	twin.py · TwinState · fidelity.py · sim.py · simulate() · harness.py · llm.py · agent.py · cli.py · load_sources()	`src/supply_twin/` · classes and functions	-	✓

ID	Shown	Source · key	Calculation	Check
F-18	28 trucks	outputs/data_quality.json · trucks_in_source	-	✓
F-19	4,320 store-SKU rates	outputs/data_quality.json · stores x skus	36 x 120 = 4,320	✓
F-20	Data quality · data_quality.json	src/supply_twin/cli.py:49 · _write_json(out / 'data_quality.json')	-	✓
F-21	load_sources() · 4 CSV files	src/supply_twin/twin.py:74-80 · four read_csv calls	-	✓
F-22	Backtest · 13 weeks	outputs/fidelity.json · holdout_days = 91	91 / 7 = 13	✓
F-23	WAPE 17.8% / 17.8%	outputs/fidelity.json · wape_store_sku_week.twin	x 100	✓
F-24	σ = 0.104 demand error	outputs/fidelity.json · rate_error_sigma	-	✓
F-25	bias −3.3%	outputs/fidelity.json · bias.twin	x 100	✓
F-26	91 days · sell 21:00 · trucks 05–19 h	src/supply_twin/sim.py:27-28, :60 · SHIFT_START, SHIFT_END, SALES_HOUR, horizon_days	-	✓
F-27	≈0.1–0.2 s per simulated quarter	docs/adr/0001-paired-seeds-and-a-practical-floor-decide-strategies.md:21 · 'runs in about 0.1-0.2 s'	-	✓
F-28	1,930 runs	outputs/search_neighbour_promo.json · simulation_runs_total (Evaluator.runs: unique runs executed, including the evaluation table)	-	✓
F-29	pool ≤ 4	src/supply_twin/harness.py:107 · min(4, os.cpu_count())	-	✓
F-30	180 s timeout	src/supply_twin/llm.py:25 · timeout: float = 180	-	✓
F-31	accept if cost CI < −0.5% and fill loss < 0.2 pts	src/supply_twin/harness.py:29, :34, :76-86 · MIN_COST_GAIN_SHARE, FILL_NON_INFERIORITY	-	✓
F-32	30 hold-out seeds × 3 scenarios	outputs/search_neighbour_promo.json; outputs/search_neighbour_promo_robustness.csv · holdout.n_seeds; robustness rows	-	✓
F-33	round 8 · 71 tested (counts up through rounds 1-8)	outputs/search_neighbour_promo.csv / .json · rounds, candidates_tested, cumulative per round	cumulative tested per round = 8, 17, 26, 35, 44, 53, 62, 71	✓
F-34	make run ≈ 2.5 min / ~2.5 min	outputs/run_log.txt · '2:26.26 total'	146.26 s / 60 = 2.44 min, rounded to the nearest half minute	✓
F-35	3 scenarios · base · promo · outage	src/supply_twin/sim.py:71-75 · SCENARIOS keys base, promo, fleet_outage	-	✓

F-27 note: Stated in the ADR from the author's runs; not re-timed by this audit.

Assumptions: A-01, A-02, A-03, A-04, A-07, A-08

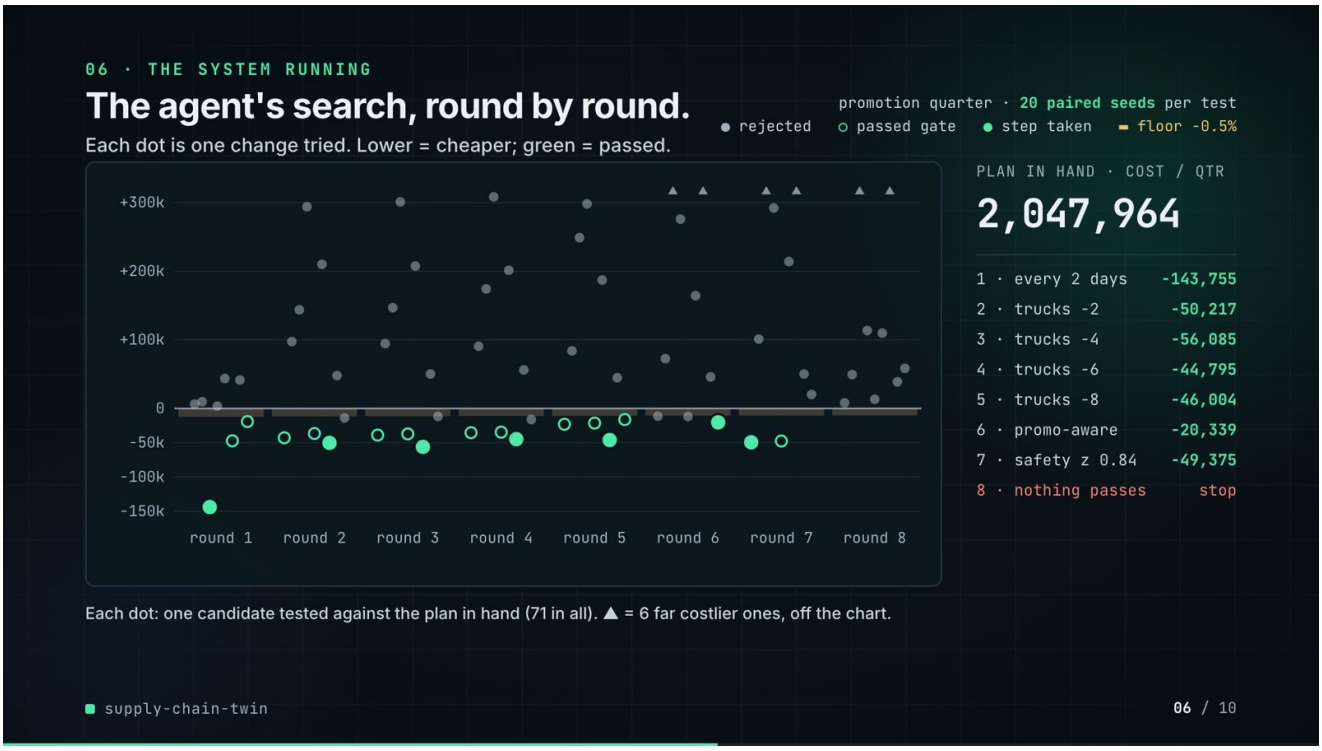
Illustrative, not data:

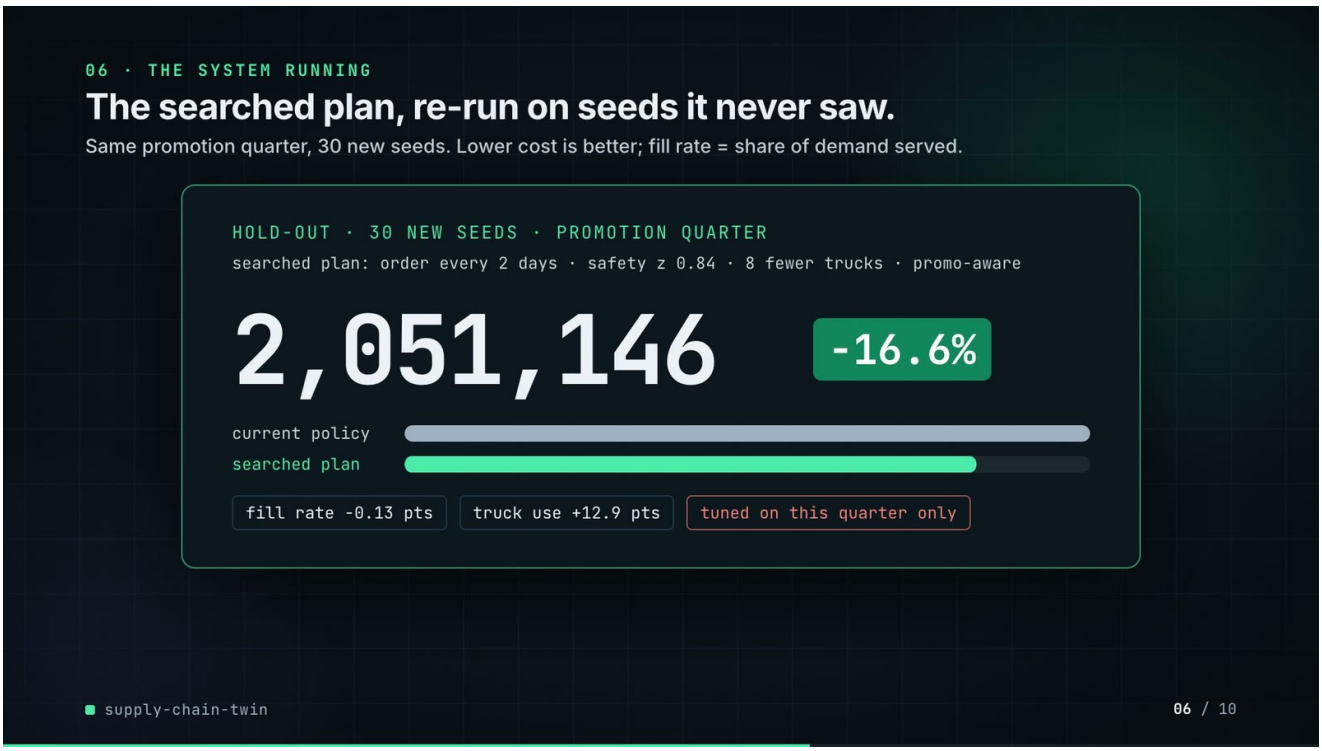
- Packets moving along the edges show execution order, not timing; the search loop is shown as 3 laps for the 8 real rounds while the counter steps through the real cumulative counts.
- The Planner box is the human decision the README describes ('a yes or no they can roll out'); the repo has no UI for it.
- 'Run cache' is the Evaluator's in-memory cache keyed by (strategy, scenario, seed) in harness.py; drawn as a store because it holds results.

To change this slide: Edit NODES / WIRES / LABELS in compositions/s03-architecture.html; F-17 checks every named module/function exists.

Slide 06: The system running

43.0-58.0 s · composition `compositions/s04-running.html (WALK block injected by tools/inject_sheet.py from assets/search_walk.json)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING
The agent's search, round by round.
Each dot is one change tried. Lower = cheaper; green = passed.
promotion quarter · 20 paired seeds per test · ● rejected ○ passed gate ● step taken ▬ floor −0.5%
Chart: cost change vs the plan in hand (−150k..+300k) by round 1..8
PLAN IN HAND · COST / QTR 2,458,533 → 2,047,964 · 1 every 2 days −143,755 · 2 trucks −2 −50,217 · 3 trucks −4 −56,085 · 4 trucks −6 −44,795 · 5 trucks −8 −46,004 · 6 promo-aware −20,339 · 7 safety z 0.84 −49,375 · 8 nothing passes stop
Each dot: one candidate tested against the plan in hand (71 in all). ▲ = 6 far costlier ones, off the chart.

(phase B) The searched plan, re-run on seeds it never saw.

Same promotion quarter, 30 new seeds. Lower cost is better; fill rate = share of demand served.

HOLD-OUT · 30 NEW SEEDS · PROMOTION QUARTER · searched plan: order every 2 days · safety z 0.84 · 8 fewer trucks · promo-aware · 2,458,347 → 2,051,146 · −16.6% · fill rate −0.13 pts · truck use +12.9 pts · tuned on this quarter only

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-16	Floor = 0.5% of quarterly cost / cost CI < −0.5% / floor −0.5%	src/supply_twin/harness.py:34 · MIN_COST_GAIN_SHARE = 0.005	-	✓
F-36	71 candidates over 8 rounds; ▲ = 6 far costlier ones	outputs/search_neighbour_promo.csv → demo assets/search_walk.json, injected into compositions/s04-running.html · round, cost_diff, accepted, chosen	dot y = cost_diff / 1000 (k); off-chart if > 320k	✓
F-37	1 every 2 days −143,755 · 2 trucks −2 −50,217 · 3 trucks −4 −56,085 · 4 trucks −6 −44,795 · 5 trucks −8 −46,004 · 6 promo-aware −20,339 · 7 safety z 0.84 −49,375 · 8 nothing passes	outputs/search_neighbour_promo.csv · rows with chosen = True	rounded to whole units	✓
F-38	2,458,533 (start of the counter)	outputs/search_neighbour_promo.csv · round 1 baseline_cost	-	✓
F-39	2,047,964	outputs/search_neighbour_promo.csv · round 8 baseline_cost	-	✓
F-40	20 paired seeds per test	outputs/search_neighbour_promo.json · search_seeds	-	✓
F-41	2,458,347 → 2,051,146	outputs/search_neighbour_promo.json · holdout.baseline_cost, holdout.cost_diff	2,458,346.6 - 407,200.4 = 2,051,146.1	✓
F-42	−16.6%	outputs/search_neighbour_promo.json · holdout.cost_diff / holdout.baseline_cost	-407,200.4 / 2,458,346.6 x 100 = -16.56%	✓
F-43	fill rate −0.13 pts	outputs/search_neighbour_promo.json · holdout.fill_diff_pts	-	✓
F-44	truck use +12.9 pts	outputs/search_neighbour_promo.json · holdout.util_diff_pts	-	✓
F-45	searched plan: order every 2 days · safety z 0.84 · 8 fewer trucks · promo-aware	outputs/search_neighbour_promo.json · final (start = z=1.28 review=1d cover=1d trucks+0 promo_aware=no)	review=2d → order every 2 days; z=0.84 → safety z 0.84; trucks-8 → 8 fewer trucks; promo_aware=yes → promo-aware	✓

Assumptions: A-02, A-03, A-04, A-05, A-08

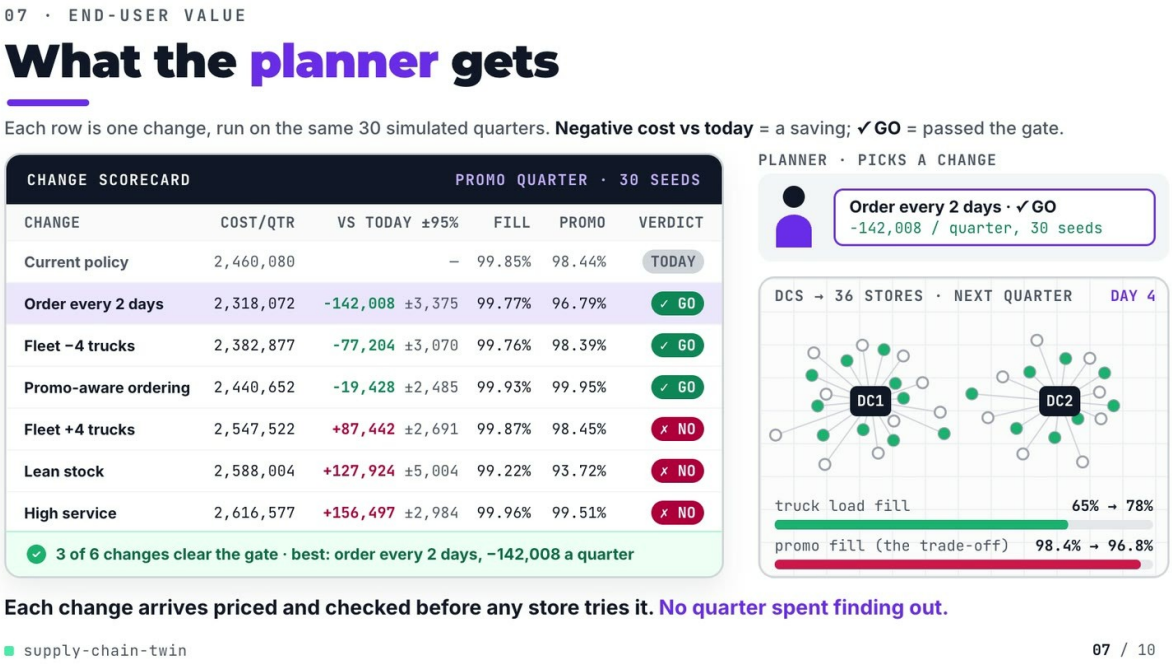
Illustrative, not data:

- Dots within a round are spread sideways only so they do not overlap; their height is the real cost change.
- The amber strips under zero are the gate's floor for that round (0.5% of the plan in hand).

To change this slide: Re-run tools/export_search_walk.py and tools/inject_sheet.py (--name WALK); layout in compositions/s04-running.html.

Slide 07: End-user value (white)

57.7-71.7 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py` from `assets/change_sheet.json`)



On-screen text (verbatim)

07 · END-USER VALUE

What the planner gets

Each row is one change, run on the same 30 simulated quarters. Negative cost vs today = a saving; ✓ GO = passed the gate.

CHANGE SCORECARD · PROMO QUARTER · 30 SEEDS — CHANGE, COST/QTR, VS TODAY ±95%, FILL, PROMO, VERDICT

Current policy 2,460,080 — 99.85% 98.44% TODAY

Order every 2 days 2,318,072 −142,008 ±3,375 99.77% 96.79% ✓ GO

Fleet −4 trucks 2,382,877 −77,204 ±3,070 99.76% 98.39% ✓ GO

Promo-aware ordering 2,440,652 −19,428 ±2,485 99.93% 99.95% ✓ GO

Fleet +4 trucks 2,547,522 +87,442 ±2,691 99.87% 98.45% ✗ NO

Lean stock 2,588,004 +127,924 ±5,004 99.22% 93.72% ✗ NO

High service 2,616,577 +156,497 ±2,984 99.96% 99.51% ✗ NO

✓ 3 of 6 changes clear the gate · best: order every 2 days, −142,008 a quarter

PLANNER · PICKS A CHANGE: Order every 2 days · ✓ GO −142,008 / quarter, 30 seeds

DCS → 36 STORES · NEXT QUARTER · DAY 1..4 · truck load fill 65% → 78% · promo fill (the trade-off) 98.4% → 96.8%

Each change arrives priced and checked before any store tries it. No quarter spent finding out.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-46	Current policy 2,460,080 … High service 2,616,577 (+156,497 ±2,984)	`demo assets/change_sheet.json` (re-run of the repo's evaluation, seeds 1000-1029), injected into `compositions/value.html`; equals `outputs/strategy_table.csv` `scenario=promo` · total_cost, cost_vs_current, cost_vs_current_ci_half, fill_rate, promo_fill_rate, beats_current	-	✓
F-47	3 of 6 changes clear the gate	`outputs/strategy_table.csv` · scenario=promo, beats_current	-	✓
F-48	best: order every 2 days, −142,008 a quarter	`outputs/strategy_table.csv` · scenario=promo, strategy='order every 2 days', cost_vs_current	-	✓
F-49	65% → 78%	`outputs/strategy_table.csv` · scenario=promo, load_fill (0.6461, 0.7760)	x 100, whole percent	✓
F-50	98.4% → 96.8%	`outputs/strategy_table.csv` · scenario=promo, promo_fill_rate (0.9844, 0.9679)	x 100, 1 decimal	✓
F-51	PROMO QUARTER · 30 SEEDS / 30 seeds	`demo assets/change_sheet.json` · n_seeds	-	✓

ID	Shown	Source · key	Calculation	Check
F-52	DCs → 36 stores, deliveries on alternating days	`data/mms_locations.csv; src/supply_twin/sim.py:291 ((day + s) % review_days)` · store_id, dc_id, distance_km	radius = 26 px + km x 0.68 px; angle illustrative	✓

Assumptions: A-02, A-03, A-04, A-05, A-06

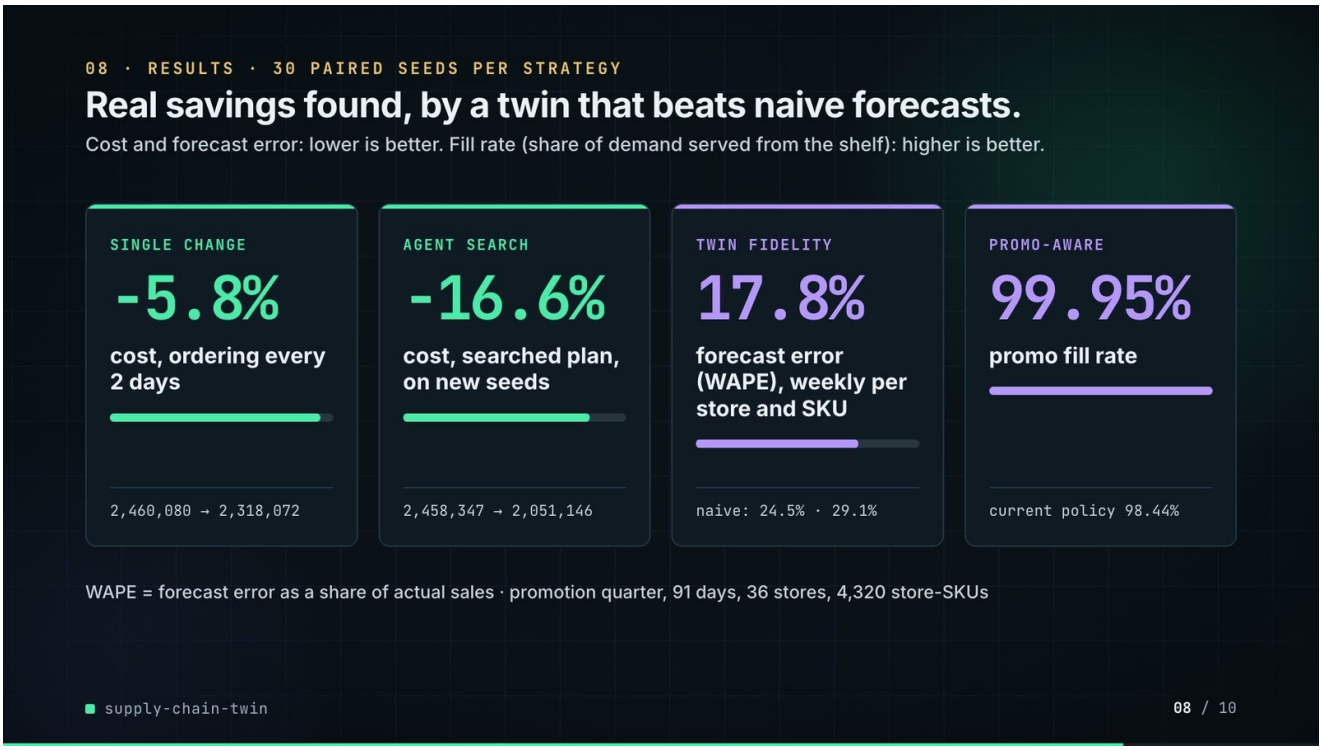
Illustrative, not data:

- Planner figure is a placeholder (no real person).
- Store angles around each DC are evenly spaced (the data has no coordinates); each store's distance from its DC is real.
- Purple trips and green dots replay the review stagger of 'order every 2 days' (which half of the stores orders each day); they are not simulated truck movements.
- 'No quarter spent finding out' is framing from the README problem statement, not a measured result.

To change this slide: Re-run tools/export_change_sheet.py, then tools/inject_sheet.py (--name SHEET); CONFIG.pick chooses the row the planner hands over.

Slide 08: Results

71.4-80.2 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · 30 PAIRED SEEDS PER STRATEGY

Real savings found, by a twin that beats naive forecasts.

Cost and forecast error: lower is better. Fill rate (share of demand served from the shelf): higher is better.

SINGLE CHANGE −5.8% cost, ordering every 2 days · 2,460,080 → 2,318,072

AGENT SEARCH −16.6% cost, searched plan, on new seeds · 2,458,347 → 2,051,146

TWIN FIDELITY 17.8% forecast error (WAPE), weekly per store and SKU · naive: 24.5% · 29.1%

PROMO-AWARE 99.95% promo fill rate · current policy 98.44%

WAPE = forecast error as a share of actual sales · promotion quarter, 91 days, 36 stores, 4,320 store-SKUs

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-04	36 stores	`outputs/data_quality.json` · stores	-	✓
F-19	4,320 store-SKU rates	`outputs/data_quality.json` · stores x skus	36 x 120 = 4,320	✓
F-23	WAPE 17.8% / 17.8%	`outputs/fidelity.json` · wape_store_sku_week.twin	x 100	✓
F-41	2,458,347 → 2,051,146	`outputs/search_neighbour_promo.json` · holdout.baseline_cost, holdout.cost_diff	2,458,346.6 - 407,200.4 = 2,051,146.1	✓
F-42	−16.6%	`outputs/search_neighbour_promo.json` · holdout.cost_diff / holdout.baseline_cost	-407,200.4 / 2,458,346.6 x 100 = -16.56%	✓
F-53	−5.8%	`outputs/strategy_table.csv` · scenario=promo, total_cost: 2,318,072 vs 2,460,080	(2,318,072 / 2,460,080 - 1) x 100 = -5.77%	✓
F-54	2,460,080 → 2,318,072	`outputs/strategy_table.csv` · scenario=promo, total_cost	-	✓
F-55	naive: 24.5% · 29.1%	`outputs/fidelity.json` · wape_store_sku_week.recent_8wk_mean, same_day_last_year	x 100	✓
F-56	99.95% · current policy 98.44%	`outputs/strategy_table.csv` · scenario=promo, promo_fill_rate	x 100, 2 decimals	✓

Assumptions: A-04, A-05, A-07, A-09

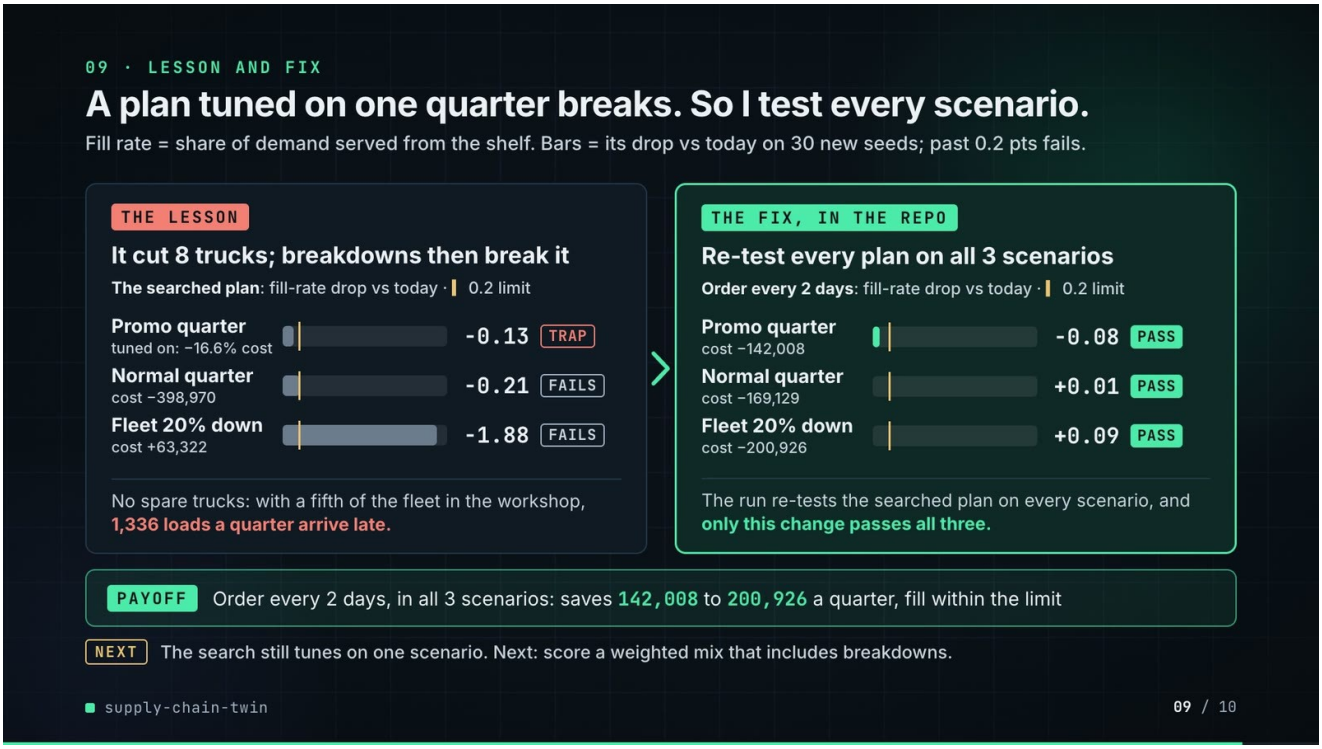
Illustrative, not data:

- Mini bars under each number are drawn to the ratio after / before (card 3: twin / 8-week mean).

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: Lesson and fix

79.8-88.8 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · LESSON AND FIX

A plan tuned on one quarter breaks. So I test every scenario.

Fill rate = share of demand served from the shelf. Bars = its drop vs today on 30 new seeds; past 0.2 pts fails.

THE LESSON · It cut 8 trucks; breakdowns then break it · The searched plan: fill-rate drop vs today · 0.2 limit · Promo quarter (tuned on: −16.6% cost) −0.13 TRAP · Normal quarter (cost −398,970) −0.21 FAILS · Fleet 20% down (cost +63,322) −1.88 FAILS

No spare trucks: with a fifth of the fleet in the workshop, 1,336 loads a quarter arrive late.

THE FIX, IN THE REPO · Re-test every plan on all 3 scenarios · Order every 2 days: fill-rate drop vs today · 0.2 limit · Promo quarter (cost −142,008) −0.08 PASS · Normal quarter (cost −169,129) +0.01 PASS · Fleet 20% down (cost −200,926) +0.09 PASS

The run re-tests the searched plan on every scenario, and only this change passes all three.

PAYOFF · Order every 2 days, in all 3 scenarios: saves 142,008 to 200,926 a quarter, fill within the limit

NEXT · The search still tunes on one scenario. Next: score a weighted mix that includes breakdowns.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-57	Promo quarter · tuned on: −16.6% cost · −0.13 · TRAP	`outputs/search_neighbour_promo_robustness.csv` · scenario=promo: fill_diff_pts, cost_diff / baseline_cost, accepted	-407,200 / 2,458,347 x 100 = -16.6%	✓
F-58	Normal quarter · cost −398,970 · −0.21 · FAILS	`outputs/search_neighbour_promo_robustness.csv` · scenario=base	-	✓
F-59	Fleet 20% down · cost +63,322 · −1.88 · FAILS · 1,336 loads a quarter arrive late	`outputs/search_neighbour_promo_robustness.csv` · scenario=fleet_outage	late loads = final_late_loads (mean over 30 hold-out seeds, one 91-day quarter) rounded	✓
F-60	0.2 limit / past 0.2 pts fails	`src/supply_twin/harness.py:29, :79-80` · FILL_NON_INFERIORITY = 0.002 on the lower bound of the fill change	-	✓
F-61	Re-test every plan on all 3 scenarios · The run re-tests the searched plan on every scenario	`src/supply_twin/cli.py:65-82 (cmd_evaluate), :157-181 (cmd_robustness), :219-222 (called after the search)` · for ... in SCENARIOS.items(); cmd_robustness(...)	-	✓
F-65	Promo quarter cost −142,008 −0.08 PASS · Normal quarter cost −169,129 +0.01 PASS · Fleet 20% down cost −200,926 +0.09 PASS	`outputs/strategy_table.csv` · strategy='order every 2 days', scenario in promo/base/fleet_outage: fill_vs_current_pts, cost_vs_current, beats_current	-	✓

ID	Shown	Source · key	Calculation	Check
F-66	only this change passes all three	`outputs/strategy_table.csv` · beats_current per strategy across the 3 scenarios	strategies with beats_current in every scenario = {order every 2 days}	✓
F-67	saves 142,008 to 200,926 a quarter, fill within the limit	`outputs/strategy_table.csv` · strategy='order every 2 days': cost_vs_current min/max; beats_current all	-max(cost_vs_current) .. -min(cost_vs_current)	✓
F-68	The search still tunes on one scenario. Next: score a weighted mix that includes breakdowns.	`README 'Known limitations' (first bullet) and 'Insights and learning'` · 'search against a weighted mix of scenarios'; 'neither is implemented'	-	✓

F-61 note: The gate rule itself is unit-tested (tests/test_harness.py::test_acceptance_rule_on_known_cases); cmd_robustness has no unit test of its own, it is exercised by make run, which wrote the robustness CSV.

Assumptions: A-02, A-03, A-04, A-05, A-06

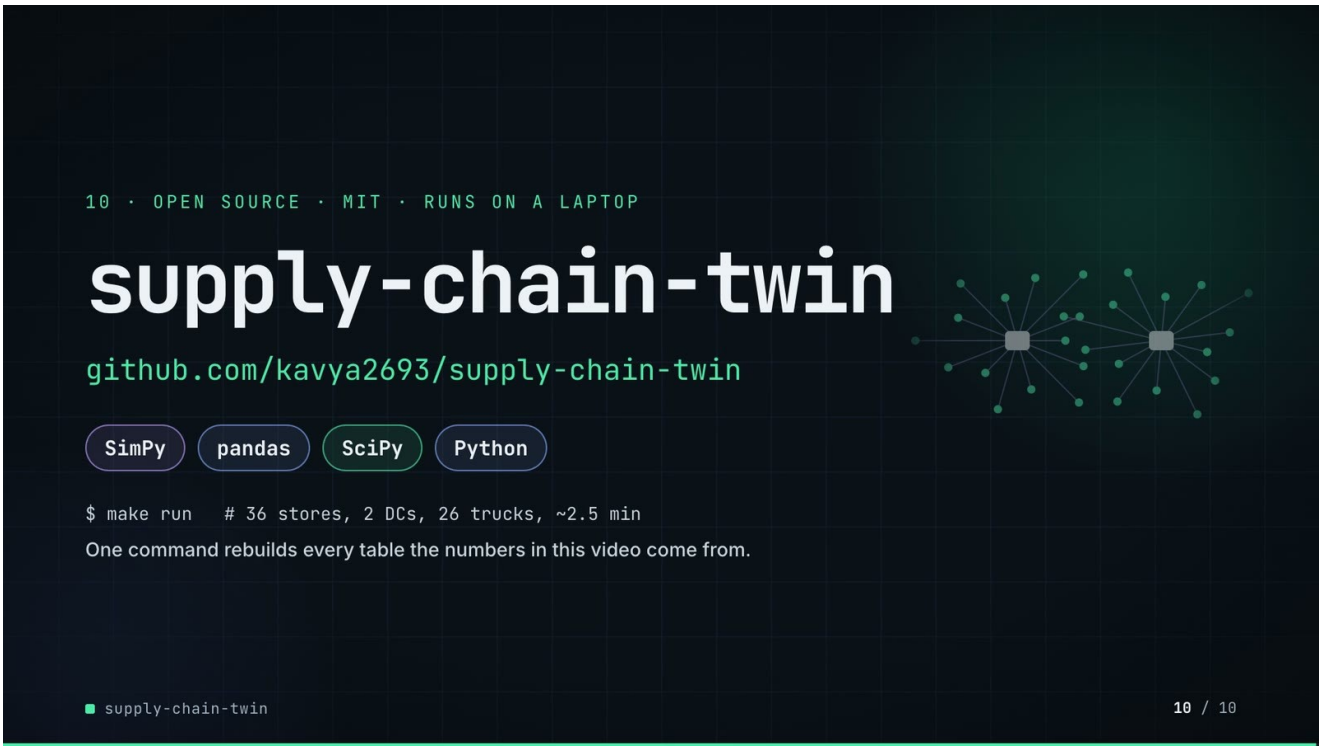
Illustrative, not data:

- Bars in both panels share one scale: 240 px = a 2.0-point fill drop (120 px per point); a gain (+0.01, +0.09) draws no bar. The amber tick in every track is the 0.2-point limit.
- Colour meaning: the searched plan's bars are neutral grey; coral marks only the TRAP label (the quarter it was tuned on, where it looks best); the fix panel, its bars and PASS chips are mint.
- The left panel uses the 30 hold-out seeds (5000-5029, robustness CSV); the right panel uses the 30 evaluation seeds (1000-1029, strategy table). Neither set was seen by the search (seeds 0-19).
- NEXT is the README's own stated fix for its biggest gap; it is not implemented.

To change this slide: Panels #twist-left / #twist-right (bar widths are inline styles on #twist-lb1..3 and #twist-rb1..3), #twist-payoff and #twist-next in compositions/s06-twist.html.

Slide 10: End card

88.4-92.8 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP

supply-chain-twin

github.com/kavya2693/supply-chain-twin

SimPy · pandas · SciPy · Python

$ make run # 36 stores, 2 DCs, 26 trucks, ~2.5 min

One command rebuilds every table the numbers in this video come from.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	2 DCs	`data/mms_locations.csv` · distinct dc_id	-	✓
F-04	36 stores	`outputs/data_quality.json` · stores	-	✓
F-05	26 trucks	`outputs/data_quality.json` · trucks_dispatchable	28 in source - 2 not dispatchable	✓
F-34	make run ≈ 2.5 min / ~2.5 min	`outputs/run_log.txt` · '2:26.26 total'	146.26 s / 60 = 2.44 min, rounded to the nearest half minute	✓
F-62	MIT	`LICENSE line 1` · -	-	✓
F-63	SimPy · pandas · SciPy · Python	`pyproject.toml dependencies` · simpy, pandas, scipy	-	✓
F-64	$ make run	`Makefile` · run target	-	✓

Assumptions: A-09

Illustrative, not data:

- The faded network on the right is decoration drawn from geometry, not data.

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, written by a seeded generator (seed 7) shaped like the three source systems: 28 trucks, 120 SKUs, 36 stores, 2 DCs, two years of daily order lines with 4% annual growth, holidays and category promotions	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`scripts/generate_data.py`	make data; make run
A-02: Acceptance floor: a change is accepted only if the whole 95% interval of its total-cost change lies below -0	F-14, F-15, F-16, F-31, F-36, F-37, F-39, F-46, F-47, F-48, F-65, F-66, F-67	04, 05, 06, 07, 09	`src/supply_twin/harness.py:34`	make run; export_search_walk.py + export_change_sheet.py + inject_sheet.py
A-03: Fill non-inferiority: a change may lose at most 0	F-31, F-36, F-37, F-46, F-47, F-57, F-58, F-59, F-60, F-65, F-66, F-67	05, 06, 07, 09	`src/supply_twin/harness.py:29`	make run
A-04: Seeds: the search uses 20 seeds (0-19) with a Bonferroni split of alpha per round; the evaluation table uses 30 seeds (1000-1029); the searched plan is confirmed on 30 hold-out seeds (5000-5029) the search never saw	F-09, F-12, F-32, F-40, F-41, F-42, F-43, F-44, F-51	02, 03, 05, 06, 07, 08	`src/supply_twin/cli.py:189-191`	make run
A-05: Cost constants are stated, not estimated: holding 25% a year, backroom overflow 1	F-14, F-36, F-37, F-39, F-41, F-42, F-46, F-48, F-53, F-54, F-57, F-58, F-59	04, 06, 07, 08, 09	`src/supply_twin/sim.py:29-34`	make run; export_search_walk.py + export_change_sheet.py + inject_sheet.py
A-06: Scenarios: a 91-day horizon; the promotion quarter puts beverages and snacks on a 14-day promotion from day 35; the outage scenario takes 20% of each DC's trucks out of service for the whole quarter	F-09, F-26, F-35, F-59, F-61, F-65, F-66	02, 05, 09	`src/supply_twin/sim.py:60-74`	make run
A-07: Simulated demand carries the store-SKU error spread measured in the 13-week backtest (sigma 0	F-10, F-23, F-24, F-25, F-56	02, 04, 05, 08	`src/supply_twin/fidelity.py`	make run
A-08: The neighbour proposer moves one lever one grid step at a time (z, review days, cover days, fleet size, promotion awareness); the search stops when a round accepts nothing (here after 8 rounds)	F-13, F-33, F-36, F-37, F-45	03, 05, 06	`src/supply_twin/agent.py:23-29`	make run
A-09: Video rounding: money to whole units with thousands separators, percentages to 1 decimal (fill rates to 2), points to 2 decimals; '3	F-06, F-10, F-14, F-24, F-34, F-42, F-49, F-50, F-53, F-56	02, 04, 05, 06, 07, 08, 10	`the composition file named on each slide`	hyperframes render
A-10: Slide 01 (industry, everyday problems) is general retail-planning knowledge used as context	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

• The whole dataset is synthetic and the twin is estimated from data produced by the same family of assumptions it then simulates, so fidelity (17.8% WAPE) is an upper bound on what a real extract would give (README 'The data').

• −16.6% (slides 06, 08) is the searched plan on the quarter it was tuned on; it fails the fill limit on a normal quarter and fails on cost and fill with 20% of the fleet down (slide 09).

• '1,930 runs' is the number of unique simulations executed when the search finished, including the evaluation table; the robustness re-test ran more afterwards.

• '≈0.1–0.2 s per simulated quarter' is quoted from ADR-0001, not re-timed by this audit.

• 'make run ≈ 2.5 min' is the author's run_log (2:26); the README says about 3 minutes on 4 cores.

• Slide 04's three intervals compare each candidate with the plan in hand at that round of the search, not always with the original policy.

• The language-model proposer is drawn in the architecture but was only tested with stubbed replies (README 'Known limitations'); every number in the video comes from the neighbour proposer.

• Slide 01 statements are general retail-planning knowledge (A-10).

• Slide 09's fix is the re-test and the per-scenario scorecard the repo runs; the search itself still optimises one scenario at a time, so a plan that holds under breakdowns is found by the evaluation table, not by the search (README 'Known limitations'). The left panel (hold-out seeds 5000-5029) and right panel (evaluation seeds 1000-1029) use different seed sets.

• Slide 09 drops the README's other 'what didn't win' items, which the README still reports: the high-service policy has the best fill rate and loses 156,497 (also on slide 07), and the language-model proposer has no live measurement (it timed out on the author's machine).

Appendix: environment

Item	Value
Commit	`a4f50f5289cc8c298a8e842f8aadd139c609579c`
Python	3.11.7
simpy	4.1.2
pandas	3.0.6
numpy	2.4.6
scipy	1.17.1
Profile / seeds	data_generator=7, search_seeds=0-19, evaluation_seeds=1000-1029, holdout_seeds=5000-5029
Run time	outputs/run_log.txt: make run 2:26.26 total (about 2.4 min) on the author's laptop; README says about 3 minutes on 4 cores